

Dairy Smart Cup building instructions

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Dairy Smart Cup in context

Advancements in Precision Livestock Farming (PLF) technologies have facilitated the automated monitoring of livestock health, behavior, and production, offering valuable insights for both farmers and researchers. In a vast range of applications, these technologies utilize various sensors to monitor cattle at an individual level, thus allowing for early detection of health issues or changes in behavior. One known link is between water intake patterns and cattle health; changes in drinking behavior might signal the onset of illness or stress. Existing solutions for cattle hydration monitoring include scale-based solutions that measure the amount of consumed water (Insentec Roughage Intake Control, RIC2Discover, Intergado, and BioControl) and timing solutions that measure the time spent at the drinking station (GrowSafe, RumiWatch). While these innovations provide accurate data on drinking patterns and water intake, their high costs and complexity often make them impractical for widespread farm use (Tuomisto and Pölönen, 2024).

The solution described in this text resulted in Smart Drinking Device (Älyjuoma-automaatti 2022) project (HAMK, n.d.) where one of project main aims was in designing a low-cost, easy to maintain and small floor level square-meter footprint device that could be used to record individual drinking behavior of animals. The intended use was for dairy cows and field tests are made with two dairy barns in Southern Finland. We aimed to design a device that would have total component cost of 200 to 250 euros to allow commercial scaling yet still being accurate in the measurement. Easy maintenance included considering robustness of the device with parts and sensors that would require very little regular cleaning. Calibration, if needed, should be automated. Furthermore, in today's barns, the requirements of usable floor area for the cows are strict and any area occupied by drinking devices is not counted usable for the cows. Thus, any structural choices that increase the footprint will induce additional costs to the barn building.

This paper focusses on the accurate measurement of water volumes consumed by dairy cows. The presented hardware can measure each individual drinking event accurately. Through identification of the cow, the daily intake of water for every individual cow can be measured. Design is based on the use of pressure sensors with water being pumped to drinking bowl through intermediate regulating tank that is filled from the water mains through controlled valve. This design allows alternating the refill of drinking bowl and regulating tank such that the hydrostatic pressure is measured from settled water body.

Identifying the cow engaged in drinking with the device was unfortunately not yet solved during the project and, therefore, not addressed in this paper. We attempted the identification of the cow via their ear tag that has been required to contain an ISO 11784 and ISO 11785 compatible RFID chip since 2021. However, in our tests, we found that this approach would likely require structures limiting the angular space of cow orientation and increasing the floor footprint, or, possibly in addition to previous, more sophisticated higher cost antenna. As these were against the design principles, later work will proceed with other possibilities. (Hattingh and Koskela, 2025)

Before the development of described project, we had experimented with flow meters and ultrasonic distance sensor. The tested, about 100-euro flow meter was found to have significant amount of noise if installed simply on the water pipe close to a regular drinking cup with nose valve likely due to irregular use of the nose valve varying between different cows. (Järvenpää, 2022; Tuomi, 2023) Ultrasonic sensor required larger diameters for measuring pipes and thus increasing the floor footprint of the possible setup. (Koskela et al., 2022) Its accuracy is also lower than pressure sensor and the presented setup provides several additional benefits such as higher flowrates almost independent of water pipe pressure and

possibility of introducing automated maintenance such as flushing or change of fresh water if not consumed for a while.

Hardware description

The working principle of the Dairy Smart Cup is based on measuring hydrostatic pressure in a water column accurately. As water is consumed, the water level in the column decreases, with an associated decrease in pressure. The water consumed is then replaced by sourcing water from the freshwater mains connection. This process is controlled by a small electronic control system, two valves, a pump, and two differential pressure sensors. Data is collected by the control system and appended to an Azure cloud database. The Dairy Smart Cup consists of five basic functional systems – the Regulating Tank, the sensor module, the drinking bowl, the control electronics, and the cloud database and visualizations. In the following we describe the basic information of each functional system:

The Regulating Tank: The working principle of the Dairy Smart Cup is based on measuring the hydrostatic pressure in a water column accurately. A high accuracy differential pressure sensor is used to measure the pressure of a water column in a normal PVC wastewater pipe with 110 mm diameter. The pipe ideally has a length of approximately 1.5 meters and is mounted vertically. This is the Regulating Tank (RGT).

The RGT is fitted with both a sensor module at the bottom, and a float sensor at the top, with an overflow outlet. The float sensor at the top is situated at a height close to the point where the pressure sensor/ADC combination is saturated. The overflow outlet directs water to floor area where it is drained in a normal process.

The sensor module at the bottom of the RGT is fitted inside a standard 110mm wastewater endcap, and the components sealed with a food-safe silicone potting compound. The components housed in the sensor module are a

- high volume pump (typically 20-30 liters per minute),
- the bottom float sensor (reed switch),
- two differential pressure sensors (range -100 to 100 mbar),
- a hose connector from the pump towards the drinking bowl,
- and a hose connector towards the water mains.

The potting compound is required to create a watertight seal for the components, and to provide rigidity for components such as the water pump. The sensor module is connected to the water mains via a 12V water electromechanical valve.

The drinking bowl is a standard stainless drinking bowl with a nose activated valve, found in many farm hardware outlets. For this project, the nose activated valve had not been removed to facilitate drinking even if the Dairy Smart Cup would be out of order. The nose valve was, however, isolated with a manual shut-off ball valve during normal operation and the barn staff was trained to use the valve if needed. A hole was drilled in the bottom of this bowl, and a brass tank feedthrough was fitted and sealed watertight. The RGT was connected to the bowl via a braided flexible PVC pipe and brass fittings to the tank feedthrough component. An additional T-piece was fitted between the brass connector of the bowl and the hose, and the plugged with a standard brass plug. An M6 screw hole is tapped into the brass plug and fitted with a Luer-Lock connector and sealed. A small transparent PVC plastic pipe connects to the second differential pressure sensor.

The purpose of the pressure sensor in the bowl was threefold:

- It measured the final level in the bowl after a drinking event, which is added to the final water volume consumed by the animal.
- Perturbations on the pressure measurement in the bowl was used as indicator that an animal had started the drinking water.

- An estimate of the (negative) change in water level was added to the bowl after the drinking event, to prevent unnecessary overflowing when the bowl is filling.

The electronic control system is based on an Arduino Mega (Pro) with the Atmega 2560 processor. A focused printed circuit board manages the functionality required on the system. The system is powered via an ATX (PC) power pack which provides both 12V and 5V at ample current (the pump and gravity feed valves requires high current at 12V). Cloud connectivity is achieved by a ESP-01 WiFi unit. Switching the high current components is achieved by using a TIP121 (npn) MOSFET. Industrial circular connectors are used to connect the ATX within the electronics enclosure, as well as the various sensors (in the Sensor Module), valves and pump.

Cloud connection is achieved via the ESP-01 WiFi module, connected to the PCB via a 2-row 8 pin header.

A visualization panel comprised of a small annotated pcb with programmable LED's gives on-site feedback on the status of the system to barn staff. The status of the RGT volume, pump and valve actuations, and the general system is indicated.

The data is captured via a high-capacity API in an Azure database. A small test mobile app was developed, whilst most of the visualization was achieved via an instance of ReDash hosted on the organization's cloud services.

A typical drinking event would therefore be registered and managed by the system in the following way:

- The start of a drinking event is signaled by a pressure perturbation in the bowl.
- The level of the bowl is monitored more closely, a level change is calculated and
- The calculated volume of water used in the bowl is replaced from the RGT by actuating the pump and opening the gravity feed valve.
- This process is repeated until the animal has stopped drinking, and
- the bowl is then refilled to a required minimum level.
- Critical data is recorded at each step and added to the Azure database.

A special feature of the system is self-calibration. Self-calibration is a way to ensure the accuracy of the system over time without the need for maintenance or manual intervention. It also serves as a very indicator of a failing pressure sensor (the heart of the system), or water leaks that may have developed. Self-calibration of both the bowl and the RGT sensors are achieved in the following way:

- **RGT Calibration:** The float sensor at the top of the RGT is mounted in such a way that it is slightly below the saturation column height of the differential pressure sensor. When the RGT is calibrated, water is let in from the freshwater mains, overflowing the float sensor position. Water is then slowly discharged from the RGT using the gravity feed valve only until the reed switch changes status. Since all the physical parameters are known, the pressure indication should be very close to a constant value. Sensor drift can be observed over time and accommodated in the measurement if the temporal drift is not too large.
- **Bowl calibration:** the bowl pressure sensor can be calibrated by slowly overflowing the bowl, at which time a constant pressure is observed. The volume of water used in this calibration is relatively small, amounting to 2-3 liters of water once every two days. The maximum pressure in the bowl is recorded and used as a system constant when calculating water volume consumed by the animal.

Build instructions

Phase 1. Construction of the regulating tank

The regulating tank is constructed from the following components:

Due to the 5V supply on the ADC (ADS1115TRG), the maximum column height would be 1.02 meters. Any sensor output more than 5V would result in a saturated reading. An additional 0.30 meters is added to allow easy installation of components and the sensor module.

- Cut the pipe to length (1.35 meter). Note that the broadened connection sections are not used – we need pipe with a consistent diameter.
- Fit a two-core cable wire to the float sensor and seal properly using an epoxy filled heat shrink tube. Allow enough length in the cable to ensure that the cable will reach the electronics enclosure (approximately 1.5 meters).
- Drill a hole in the steel bracket to fit the float sensor. Fit the float sensor upright onto the bracket (when in the upright position, the reed switch is not activated if the water level is below the sensor).
- Fit the wiring of the
- Measure about 1.2 meter from the bottom of the pipe (0.12 meter for the sensor module at the bottom, and 0.8 meter to ensure ADC saturation when the float sensor is reached). Install the stainless-steel bracket at this point. Fasten with stainless machine screws and nuts, using a sealant to ensure water integrity.
- Use a wastewater endcap and drill a hole in the flat part, ensuring that the plastic elbow fits tightly in the hole. Attach two plastic elbows with a small length of translucent plastic pipe so that the second elbow hangs over the edge if the first elbow is fitted in the hole. Fasten with a bracket. Add the rest of the translucent PVC pipe (it can be tied to the RGT with a cable tie for now).
- Fit the two wastewater pipe double extenders to the ends of the pipe. Make sure that the internal seals inside the extenders are intact before you attempt this step.

Phase 2. The electronics

Building the electronics can be achieved following these steps:

- Have the following circuit boards manufactured:
 - Main board
 - Pressure sensor board
 - Infoboard
- Order the electronic components and cables according to the BOM.
- When soldering and assembling the circuit boards, it may pay off to observe general rules and precautions:
 - Adequately ventilating the work room
 - Make sure the components are correctly orientated
 - Antistatic precautions
 - Follow the component datasheet instructions in terms of soldering and temperature envelopes
 - Testing as you progress to prevent long delays later looking for problems
 - Some soldering is more difficult – you might need specialized equipment for the small SMD footprint components.
 - General safety rules when working in an electronic workshop
- Build the main board first, testing regularly to ensure that no mistakes have been made.
- Create internal cables for power management and managing the high current components. In our design we used from discarded heavy brass cable electrical works in the building.
- Create internal cables for the sensor modules – small diameter cables are needed to connect the JST headers to the panel mounted circular connectors. Use a JST crimp tool to make up the cables. Make sure the wires are correctly positioned in each JST housing (and that it corresponds to the header orientation on the board).
- Test all the connections and headers and test the system for functionality. It is suggested that the sensor module is tested for all functionality after assembly, just before the potting compound is added (See the end of Phase 3).
- Specific tests to run:

- Test the ADC's and pressure sensors to ensure that it functions as planned (fitting a plastic small plastic pipe on all four orifices and blowing them should produce a measurable reading)
- Test that the reed switches in the float sensors are working properly
- Test that the pump starts correctly (take care not to run it for more than a second or two without water)
- Test that the two electromechanical valves open and close. Note that the mains valve requires backpressure to open properly.
- Test the ESP-01 WiFi module
- Test the API and some data transfer to see if the ecosystem functions as planned.

Phase 3. Assembly of the sensor module

The sensor module is the most important part of the build, and care needs to be taken to follow the steps accurately, since at some point the changes are permanent and the module is fixed. The sensor module is constructed from the following components:

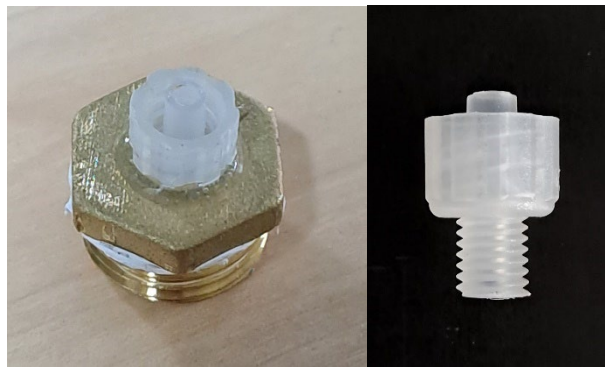
The sensor module is constructed in the following way:

- Prepare the pressure sensors. The sensors need to be soldered to the PCB's designed for this purpose. Make sure the sensors and ADC's work as required before taking the next steps.
- Prepare a cable ring by winding some of the 3mm diameter pipe around and push the ends through the holes provided (the larger diameter holes close to the center column). The sensors need to be inserted in the hollow part of the center column (do this in a later step).
- Cut a disk (approximately 5mm thick) from the nylon rod. Drill four holes that correspond with the spacing on the pressure sensor barbs. The holes need to be as close as possible to each other. Two holes will be left open to atmosphere, and the two other holes will be connected to the RGT and bowl pressure pipe eventually.
- The hole corresponding to the bowl pressure measurement needs to be tapped to fit the Luer Lock fitting. Fit a 1mm diameter pipe to both the measurement barbs of the sensors. One pipe will be used to measure pressure in the RGT, and the other will be used to measure pressure in the bowl.
- Fit the pressure sensor barbs with the 1mm inner diameter pipe and push the pipe through the corresponding holes in the nylon disk.
- Drill three holes in the end-cap to position the pump, tank feedthrough and the pressure sensor disk. Leave enough space to install the pressure sensor disk and make the construction watertight.
- Drill a hole in the end-cap floor to fasten the bottom float sensor. This can also be achieved by dipping and supporting the sensor whilst the potting compound settles (therefore without drilling any holes).
- Prepare the data cables. A length of nine-core cable is required. Make provision for enough length to reach from the sensor module to the electronics enclosure. A length of one meter should be sufficient. It is a good idea to purchase either a color coded or numbered cable. The cores in the cable need to be soldered to the 9-pin circular connector. Make a note of the corresponding pin positions and the core numbers (or colors). Using a multimeter, test the cores for continuity. (Important, the panel mount receptacle needs to be soldered, and installed prior to this step. If it has been done, and the connections have been made inside the electronic enclosure, the system could be fully tested before the potting compound is added).
- Secure the pump in the end-cap, using an epoxy glue.
- Fit the tank feedthrough and apply some epoxy to seal the connection. Glue a small 15mm piece of pipe to provide a hole from the tank feedthrough to the top of the end-cap (to provide a water connection from the feed-through to the eventual top of the potting compound).
- Ensure that all the elements are working, and that elements are properly sealed. Fill the end-cap with potting compound (enough to provide structural strength to the elements inside, as well as providing watertight sealing). This step is permanent and once the potting compound has settled, no changes are possible!

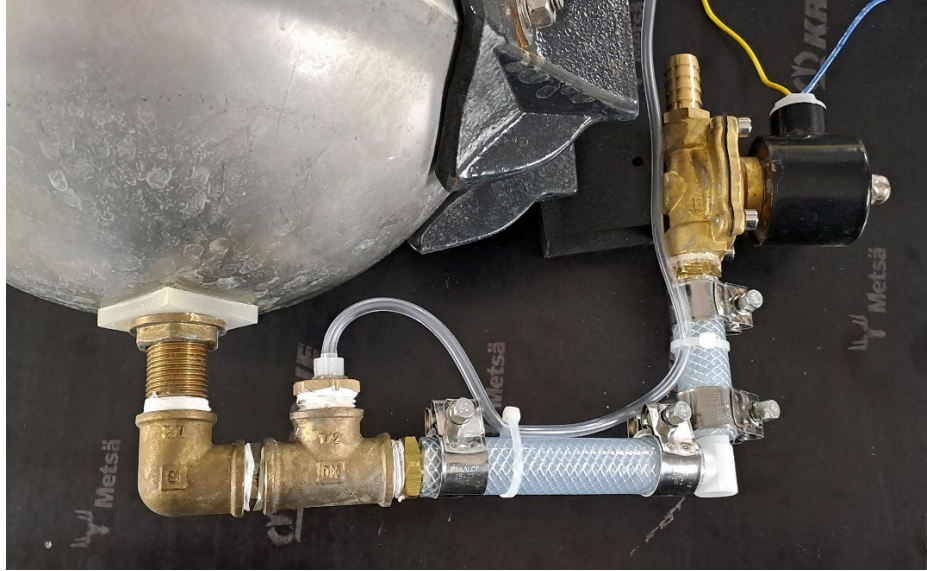
Phase 4. Assembly of the bowl

The final stage before installation is assembling the bowl components and assembling the complete machine:

- Cut a small washer from a thin rubber mat (not more than 2mm thickness) to fit below the tank feedthrough flange. Cut another 5mm disk from the 50mm diameter nylon rod and drill a hole into the disk (the diameter must be large enough for the tank feedthrough to push through).
- Drill a hole into the bottom of the bowl and fit the tank feedthrough in the hole. Remember to fit the rubber washer before pushing the feedthrough through the bowl. Fit the nylon disk between the bowl bottom and the feedthrough nut and fasten the nut. This step is important to ensure that the nut locks properly below the bowl metal – the metal may be too thin for the feedthrough to seat properly.
- Attach the elbow bend (using Teflon tape to seal) to the feedthrough.
- Attach and screw on a brass barrel nipple.
- Drill a 5.4mm hole in the brass stop, tap a 6mm thread into this hole. Be careful not to strip the thread – the metal on the brass plug top is thin. Screw a Luer Lock fitting into place and seal with an epoxy glue. Let it cure.



- Attach and tighten the brass top to the brass T-piece.
- Attach and tighten the T-piece onto the elbow fitted before
- Attach a hose connector (19mm) to the open end of the brass T-piece.
- Measure the distance from the hose connector on the bowl to the wall – this should be about 24cm. Cut a length of the braided plastic pipe to fit onto the hose connector and secure with a pipe clamp.
- Use a plastic elbow (19mm) and attach to the end of the pipe, again securing it with a pipe clamp.
- Fit a brass hose connector to the outlet side of the gravity feed valve. Measure a small length of the braided plastic pipe (about 10cm) and connect to the valve hose connector and the plastic elbow. Secure both ends with a pipe clamp.
- Fit another brass hose connector to the inlet side of the gravity feed valve. The assembly should look like this at this stage:



- Decide on a suitable height at which the RGT will be installed. As a suggestion backed by experience, the bottom of the RGT should be well out of reach of cows, since they will investigate and damage the cables and pipes from this fixture. In my experience, this is typically in excess of 2.3 meters. Measure the distance between the sensor module (pump exit hose connector) and the bottom of the bowl – and cut a length of braided pipe. This pipe will be fitted to the pump hose connector and the gravity feed valve inlet, but not yet.

Phase 5. Installation

Prior to installation, ensure that a qualified plumber has prepared a flange and ball shut-off valve where you can connect to the water mains. Also ensure that a qualified electrician has installed an electrical mains plug where the ATC power supply can be plugged in. The latter is a critical safety issue – make sure that the setup is electrically secure and safe – do not hesitate to ask the local electrician for advice. Once these two activities are finished, you can proceed with the installation:

- Fasten the RGT to the structures you have identified.
- Fasten the bowl to the permanent structures in the barn at a comfortable height for cows to reach.
- Prepare a length of braided pipe and two standard threaded couplings – connect and seal properly. On the outlet side of the valve, connect the threaded coupling to the tank feedthrough on the Sensor Module. Seal it appropriately. Attach a hose connector to the inlet side of the valve, measure the distance between the valve and the shut-off ball valve on the mains, and cut a length of the braided plastic pipe. Fit a hose connector to the shut-off valve, fit the hose and secure with high quality pipe clamps. (For a more permanent installation, the braided hose can be replaced by brass or stainless-steel piping by a plumber)
- Attach the RGT (pump hose connector) and bowl valve (hose connector on the inlet) using the long length of braided pipe measured and cut before. Secure both ends with pipe clamps.
- At this point, it is important to check all the fittings before the mains shut off valve is opened.
- Open the mains valve and check the system for any leaks. Fix any leaks before carrying on. If possible, actuate the mains valve with a 12V battery or power source to fill the RGT and check for leaks. Areas to pay special attention to are the pump location, the braided pipe from the pump hose connector to the gravity feed valve, the tank feedthrough connector, and the pressure sensor and cable entries into the sensor module (leaks here are unlikely, but it is important to check). Also check that the rubber seals internal to the wastewater pipe double sleeve extender functions as required. There should be no droplets forming anywhere on these connections. (It is advisable to have some paper towels with you in the barn to dry any water from these surfaces before you actuate the mains valve).

- Fasten the electronics enclosure and ATX power supply in the selected spot. We fixed it to a marine ply board out of reach of cows.
- Connect the ATX power supply, the sensor module, and the high current connector (pump, gravity feed valve, mains valve) to the electronics enclosure. Double check all connections! Note that the braided pipe from the pump to the gravity feed valve is likely to fill with water at this stage – the pump does not represent water barrier.
- If all the surfaces and connections are free of leaks, it is time to switch the system on. Plug the ATX power cable into the mains socket.
- The system will respond in the following way:
 - An internet connection will be made via WiFi, when this is ready,
 - The system will test how full the RGT is and top it up to float sensor.
 - The first RGT pressure sensor calibration will now be performed. Once this is successful, the bowl will be filled water from the RGT. A bowl pressure calibration will now be performed.
 - Once the bowl pressure calibration is completed, the system will fill the RGT.
 - The system is now ready.
 - Note that specific actions will be displayed on the info board. If all the LED's are green, the system is ready and operational. You can test the system by scooping water from the bowl, noting that the RGT is filled again after the actions are completed (and all LED's are green again).

Operation instructions

The unit is fully automated, and should run without any manual interventions, apart from the initial installation and adjustments. There are tasks that needs to be performed regularly by barn staff. The bowls needs to be cleaned in the normal way (this may differ between barns due to different standard operating procedures).

Acknowledgments

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